

Qspiders Internship Report

Summer Training Report-2 : ES-453

submitted in partial fulfillment of the requirement
for the award of the degree of

Bachelor of Technology in Computer Science & Engineering

Submitted by
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Nov/Dec 2025

DECLARATION

This is to certify that the material embodied in this Internship Report titled “**MERN Development Internship at QSpiders**” being submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is based on my original work. It is further certified that this internship work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma. My indebtedness to other works has been duly acknowledged at the relevant places.

Shobhit Shukla

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CERTIFICATE FROM THE COMPANY

This is to certify that Shobhit Shukla (Enrollment No. 04618002722), a student of Delhi Technical Campus, affiliated to Guru Gobind Singh Indraprastha University, has successfully undertaken an internship at QSpiders, Noida Sector 16, from 8th August 2025 to 8th November 2025.

During this period, he worked under the supervision and guidance of Mr. Shambhu Kumar, and actively participated in the MERN Development internship program. He contributed to various full-stack development modules, including frontend development using React.js, backend API creation using Node.js and Express.js, and database integration with MongoDB.

He has shown dedication, professionalism, and a strong technical understanding throughout his internship tenure.

Deeksha

HR, Qspider

CERTIFICATE FROM THE INSTITUTE

This is to certify that the work embodied in this Internship Report titled “**MERN Development Internship at QSpiders**”, being submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is original and has been carried out by **Shobhit Shukla (Enrollment No. 04618002722)** under my supervision and guidance.

It is further certified that this internship work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma to the best of my knowledge and belief.

Ms. Gunjan Sethi

Designation

ACKNOWLEDGEMENT

First and foremost, I am deeply grateful to **Mr. Shambhu Kumar**, my project supervisor (Industry), for their valuable guidance, support, and encouragement throughout this journey. Their expertise and insights were instrumental in shaping the direction of this project.

I would also like to extend my appreciation to **Ms. Gunjan Sethi** and staff of the **Computer Science and Engineering** at **Delhi Technical Campus, Greater Noida** for providing me with the necessary resources and knowledge to undertake this project.

Shobhit Shukla
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TABLE OF CONTENTS

	Page No.
Declaration	i
Certificate from the Company	ii
Certificate from the Institute	iii
Acknowledgement	iv
Abstract	v
List of Figures	vi
List of Tables	vii
	Page No.
Chapter 1: Introduction of the Company	8-9
1.1: About the Company	
1.2: Profile of the Company	
Chapter 2: Introduction of the Internship	10-11
2.1: About the Internship	
2.2: Objectives of the Internship	
2.3: Roles and Responsibilities	
Chapter 3: Internship Activities	12-15
3.1: Task Description	
3.2: Tools / Technologies / Platform used	
3.3: Technical Application	
3.4: Challenges Faced	
Chapter 4: Contribution to the Organization	16
Chapter 5: Learning and Development	17-18
Chapter 6: Summary and Conclusion	19
Chapter 7 :Bibliography	20

LIST OF FIGURES

Figure No.	Figure Title	Page No.
1.	Qspiders Company Logo	9
3.1	Home Page	12
3.2	Ride Listing Page	12
3.3	Route Comparison Page	13
3.4	Map navigation	14
3.5	GitHub Repository	14

LIST OF TABLES

Table No.	Table Title	Page No.
-----------	-------------	----------

CHAPTER 1: INTRODUCTION OF THE COMPANY

1.1 About the Company

QSpiders is one of the leading software training and skill-development organizations in India, dedicated to providing high-quality IT education and professional training. Headquartered in Bengaluru, the company specializes in preparing students and professionals for careers in software development, software testing, and modern full-stack technologies. QSpiders focuses on delivering industry-relevant, practical, and job-oriented training that equips learners with strong technical foundations and real-world project experience.

With expertise spanning Full Stack Development, MERN Stack, Software Testing (Manual & Automation), Java, Python, SQL, and corporate-level training, QSpiders helps individuals build strong development skills, understand industry workflows, and become job-ready. The organization emphasizes real-time projects, hands-on coding, placement preparation, and soft-skills development—enabling learners to align their capabilities with modern IT requirements.

During my MERN Development internship at QSpiders, I worked on building full-stack applications, understanding the architecture of modern web systems, and applying concepts such as Express routing, MongoDB integration, React components, and API development. This internship allowed me to experience hands-on coding, real-world development practices, and exposure to full-stack project workflows. My role involved implementing various functionalities, understanding backend–frontend integration, and building components aligned with MERN standards.

This training experience significantly enhanced my technical knowledge, improved my problem-solving abilities, strengthened my understanding of development processes, and helped me apply concepts to real-time tasks. It also gave me exposure to industry expectations, better preparing me for professional opportunities in full-stack development.

Website: <https://qspiders.com/>

Industry: IT Training & Software Skill Development

Type: Private Training Organization

Headquarters: Bengaluru, Karnataka, India

Team Size: 1000+ Trainers & Staff Across India

Specializations: MERN Development, Full-Stack Training, Software Testing.



(Fig 1: Qspiders Company Logo)

1. QSpiders Overview

QSpiders is one of India's leading software training and skill-development organizations, dedicated to providing high-quality IT education and professional training. Headquartered in Bengaluru, the company focuses on preparing students and professionals for careers in software development, software testing, and full-stack technologies. QSpiders emphasizes practical learning, industry-relevant training, and real-world project exposure to equip learners with strong technical foundations and job-ready skills.

2. Vision

To become a global leader in IT training by delivering practical, reliable, and industry-driven technical education that empowers students to build successful careers in the software industry.

3. Mission

To provide structured, hands-on training in emerging technologies through expert instructors, real-time applications, and a learner-centric approach—ensuring that students gain comprehensive technical expertise, analytical skills, and confidence to meet industry expectations.

4. Work Culture

QSpiders fosters a disciplined, collaborative, and growth-oriented learning environment. The institute encourages continuous practice, interactive sessions, coding challenges, real-time assignments, and consistent mentor support. Trainers guide students through structured modules, mock interviews, and project-based tasks, enhancing both technical and professional capabilities.

During my internship, I experienced QSpiders' systematic training approach, daily coding practices, and strong mentor guidance. Their structured full-stack MERN curriculum, combined with practical exercises and doubt-clearing sessions, helped me improve my technical understanding, problem-solving abilities, teamwork, and industry readiness. The exposure to real-world full-stack concepts—such as API development, MongoDB integration, and React component structuring—significantly strengthened my development skills.

CHAPTER 2: INTRODUCTION OF THE INTERNSHIP

2.1 About the Internship

The internship at QSpiders provided an excellent opportunity to gain practical exposure to Full Stack Web Development using the MERN Stack and understand real-world development workflows. The internship period extended from 8th August 2025 to 8th November 2025, during which I worked as a MERN Development Intern and actively participated in hands-on training sessions, coding assignments, and module-based project work.

The primary focus of my internship was to develop skills in building full-stack web applications using MongoDB, Express.js, React.js, and Node.js. Throughout the program, I worked on various real-time tasks such as creating REST APIs, designing responsive user interfaces, integrating backend services, managing state in React, and connecting applications with MongoDB databases. These tasks helped me understand how modern web systems are structured and how different layers of a full-stack application communicate with each other.

The internship offered a holistic understanding of full-stack development workflows, including requirement analysis, frontend and backend integration, API development, routing, database design, debugging, and deployment fundamentals. Working under the guidance of Mr. Shambhu Kumar (Industry Mentor), I learned how to apply theoretical concepts in a practical environment while enhancing my technical confidence, problem-solving abilities, logical thinking, and collaborative skills.

2.2 Objectives of the Internship

The main objectives of the internship were as follows:

- ❖ To gain hands-on experience in full-stack web application development using technologies such as React.js, Node.js, Express.js, and MongoDB.
- ❖ To understand and apply industry-standard software development practices, including version control using Git/GitHub, API development, debugging, and deployment fundamentals.
- ❖ To develop and implement functional modules as part of MERN-based full-stack applications and understand component integration across the frontend and backend.
- ❖ To enhance professional competencies such as teamwork, communication, analytical thinking, and time management through structured training and mentor-guided sessions.
- ❖ To bridge the gap between academic learning and real-world development by working on task-based assignments, coding challenges, and practical project implementation.
- ❖ To learn and follow structured training workflows while collaborating effectively with mentors and peers in a disciplined, practice-driven learning environment.

2.3 Roles and Responsibilities

As a MERN Development Intern at QSpiders, I was responsible for a wide range of technical and practice-oriented tasks that contributed to my understanding of full-stack development. My key responsibilities included:

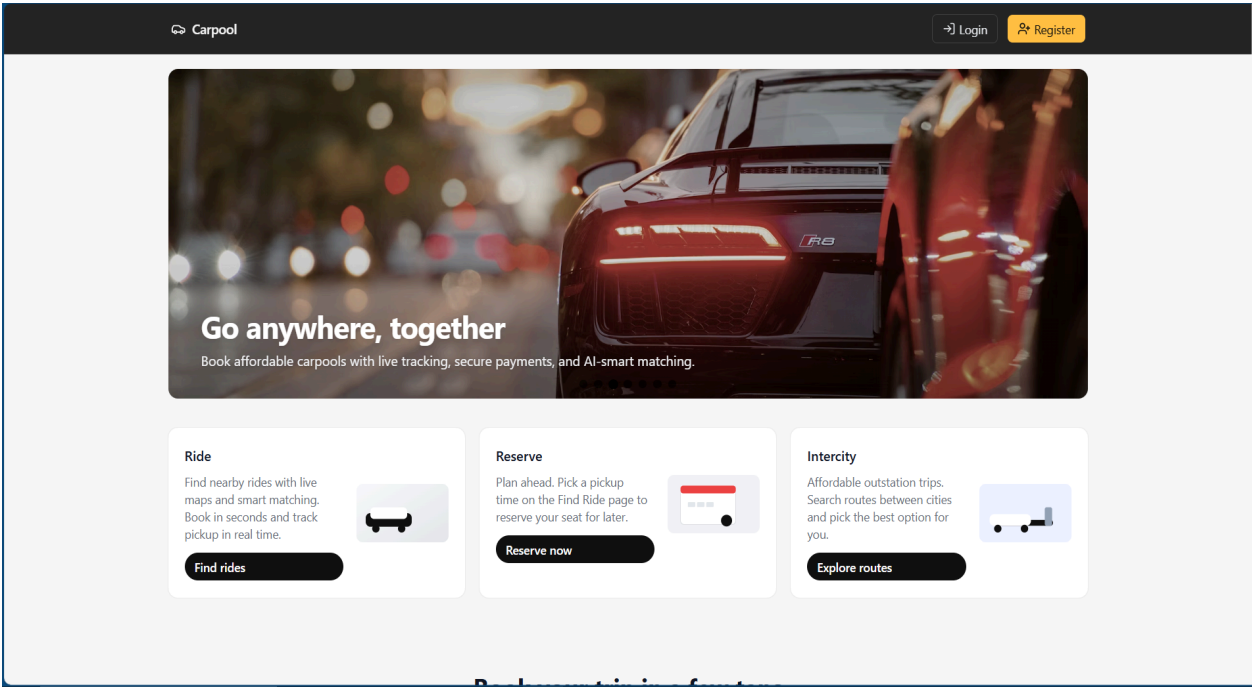
- ❖ Designing and developing front-end interfaces using React.js, focusing on building responsive, interactive, and user-friendly components.
- ❖ Building backend functionalities and RESTful APIs using Node.js and Express.js, including routing, middleware creation, and API testing.
- ❖ Working with MongoDB for data storage and retrieval, implementing CRUD operations, managing collections, and integrating database logic with backend APIs.
- ❖ Understanding and applying authentication and authorization concepts, including JWT-based authentication and secure route handling.
- ❖ Integrating frontend and backend systems, ensuring smooth data flow and consistent UI behavior through API consumption and state management.
- ❖ Collaborating with mentors and peers, participating in discussions, resolving doubts, and improving solutions based on feedback and real-world examples.
- ❖ Debugging and optimizing code, resolving issues using browser dev tools, Postman, and console-based debugging techniques.
- ❖ Managing code versions through Git and GitHub, working with repositories, commits, branches, and maintaining clean project structure.
- ❖ Participating in module-based practice sessions, weekly assessments, and progress evaluations, guided by my mentor, Mr. Shambhu Kumar.

This internship experience not only strengthened my technical expertise but also helped me develop a deep understanding of real-time development workflows, problem-solving strategies, and teamwork in a structured learning environment.

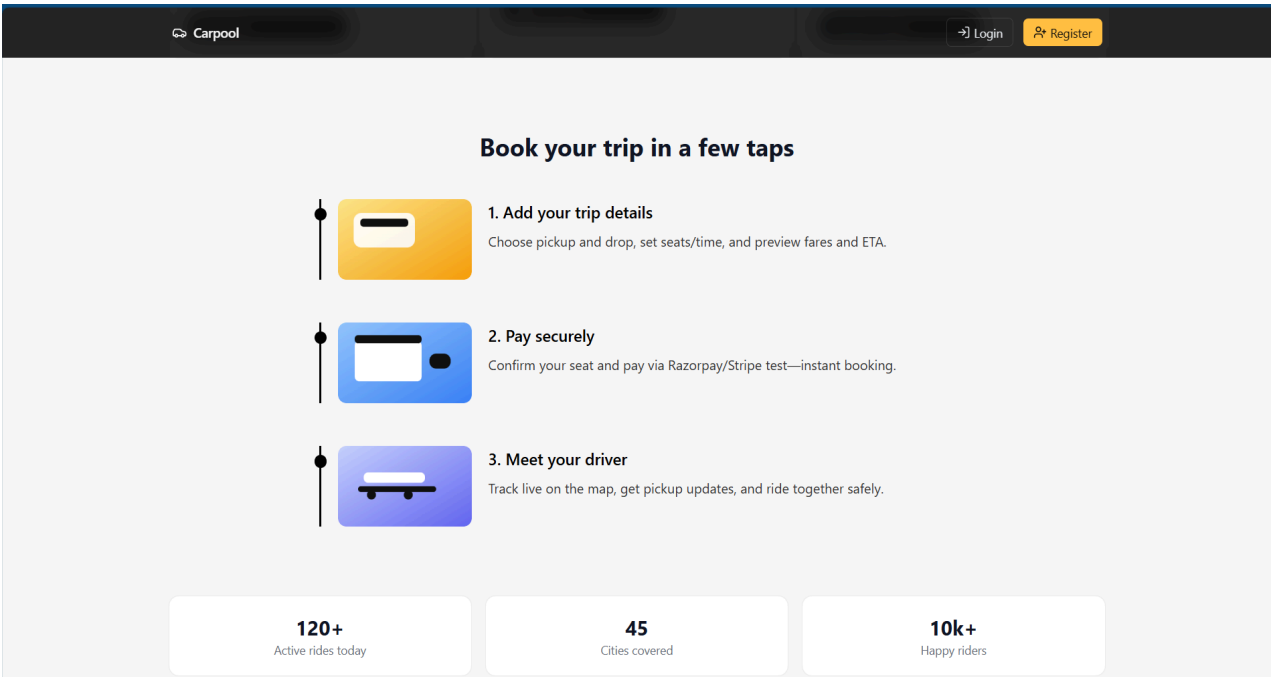
CHAPTER 3: INTERNSHIP ACTIVITIES

3.1 Task Description

During my internship at QSpiders, I actively participated in the design and development of an AI-Based Carpool System, a web application aimed at optimizing shared transportation through real-time tracking, route optimization, dynamic pricing, and intelligent ride-matching algorithms. The project’s objective was to build a modern, scalable platform that allows users to book rides, offer rides, and match with nearby travelers based on location, availability, and route similarity.



I focused on full-stack development, contributing to both frontend and backend. My work included building modules for user onboarding, dashboard analytics, partner discovery, and data visualization.



I worked closely with my mentor (Mr. Shambhu Kumar) and collaborated with other interns to ensure the platform followed modern design principles, scalability standards, and efficient database structuring. The tasks were distributed according to a structured training workflow, with continuous practice evaluations and module-wise reviews.

❖ The major tasks I performed included:

- ❖ Designing reusable frontend components using React.js and Tailwind CSS.
- ❖ Implementing backend APIs using Node.js, Express.js, and MongoDB.
- ❖ Integrating OpenRouteService APIs for live route calculations and distance optimization.
- ❖ Building ride-matching logic using AI-based similarity scoring between routes.
- ❖ Developing real-time ride updates using Socket.io for driver-rider communication.
- ❖ Creating dashboards for users, drivers, and admin to manage their activities.
- ❖ Testing & debugging across multiple device sizes for responsiveness.

3.2 Tools / Technologies / Platforms Used

The internship provided hands-on experience with a range of modern technologies, frameworks, and tools commonly used in the software industry.

Frontend Technologies

- ❖ **React.js:** For building scalable UI interfaces and managing dynamic rendering.
- ❖ **Tailwind CSS:** For responsive, utility-first styling.
- ❖ **Framer Motion:** For smooth UI animations.
- ❖ **Axios / Fetch API:** For interacting with backend APIs.

Backend Technologies

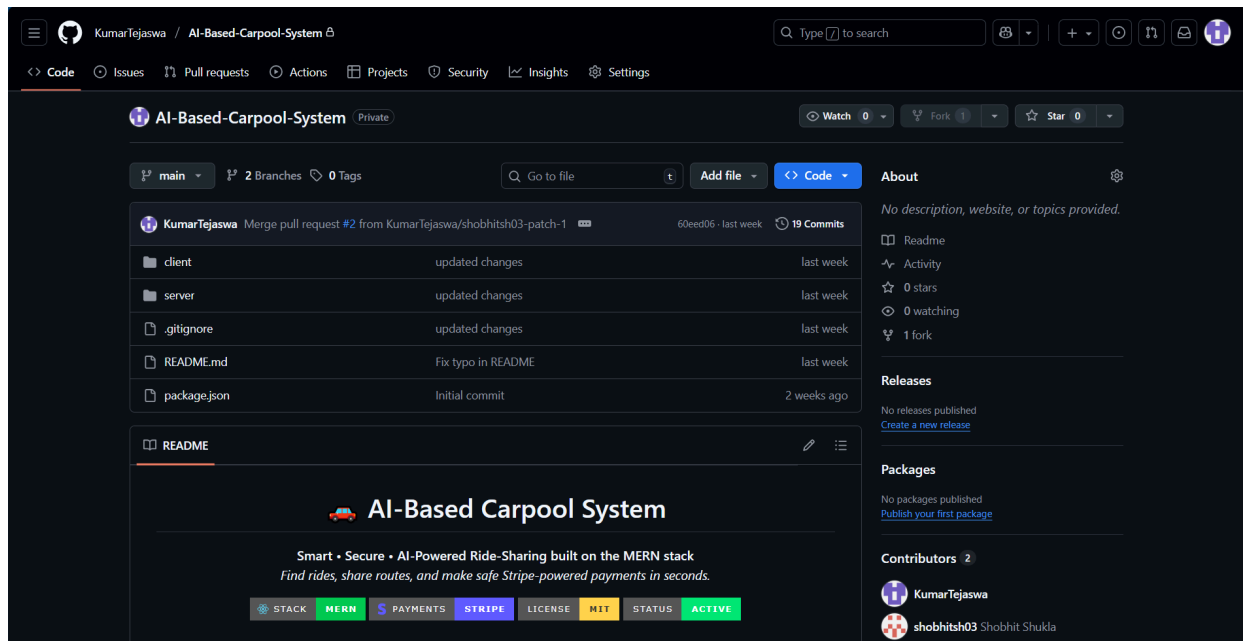
- ❖ **Node.js & Express.js:** For server-side logic, API routing, and business operations.
- ❖ **MongoDB / MongoDB Atlas:** For managing users, rides, routes, and booking details.
- ❖ **JWT Authentication:** For secure login and protected routes.
- ❖ **Socket.io:** For real-time ride updates and driver-passenger communication.

Cloud, Maps & Integration Tools

- ❖ **OpenRouteService API:** For route optimization and travel distance calculations.
- ❖ **Mapbox / Leaflet:** For map rendering within the platform.
- ❖ **Ngrok:** For temporary tunneling during testing.
- ❖ **Git & GitHub:** For code collaboration and version control.

Development Tools

- ❖ **VS Code:** Primary coding environment.
- ❖ **Postman:** For testing and validating APIs.
- ❖ **MongoDB Compass:** For database visualization and debugging.
- ❖ **Slack / Trello:** For communication and task track



3.3 Technical Application

he internship allowed me to apply full-stack development concepts to real-world scenarios. I implemented dynamic UI elements, API integrations, database interactions, and intelligent routing mechanisms.

Some of the major technical applications included:

- ❖ **Dynamic Frontend Rendering:** Creating modular UI components in React.js for maintainability.
- ❖ **AI-Based Route Matching:** Implementing route comparison logic using latitude–longitude geometry and distance scoring.
- ❖ **REST API Development:** Building and connecting APIs for authentication, ride creation, ride booking, and history management.
- ❖ **Database Optimization:** Designing efficient MongoDB schemas for ride management and search operations.
- ❖ **Responsive Design:** Ensuring UI compatibility across different device sizes.
- ❖ **Real-Time Communication:** Implementing live ride updates using Socket.io.
- ❖ **Deployment Concepts:** Understanding hosting workflows for MERN applications.

3.4 Challenges Faced

During the internship, I faced several challenges that helped strengthen my skills and adaptability:

- ❖ **Route Optimization:** Calculating shortest and most efficient routes required complex API handling.
- ❖ **Real-Time Systems:** Ensuring smooth communication between passengers and drivers was initially challenging.
- ❖ **Database Structuring:** Designing efficient collections for dynamic data such as rides and coordinates.
- ❖ **UI Responsiveness:** Maintaining consistent layout behavior on mobile and desktop screens.
- ❖ **Merge Conflicts:** Working with Git repositories helped me learn branching strategies.
- ❖ **Time Management:** Handling multiple modules within training deadlines improved productivity.

By overcoming these challenges, I strengthened my full-stack development skills, improved problem-solving abilities, and gained practical insights into real-world project development.

CHAPTER 4: LEARNING AND DEVELOPMENT

The internship at QSpiders was a transformative experience that bridged the gap between theoretical academic learning and practical industry application. Over the three-month training period, I gained valuable technical, professional, and interpersonal skills that have significantly contributed to my overall growth as a Computer Science Engineer.

Working on the AI-Based Carpool System gave me hands-on exposure to the MERN stack, API integration, route optimization algorithms, cloud-based development, and structured training workflows. This chapter summarizes the major areas of learning and development achieved during my internship.

4.1 Frontend Enhancement and User Experience Design

I developed and optimized the frontend components of the platform using React.js and Tailwind CSS, focusing on creating an intuitive and responsive interface. I implemented modular UI components, making the code more reusable and maintainable.

Additionally, I enhanced the user experience by integrating smooth transitions using Framer Motion and ensuring consistent responsiveness across devices.

4.2 Backend Development and API Implementation

On the backend side, I contributed to building and testing RESTful APIs using Node.js and Express.js. I implemented API endpoints for ride creation, booking operations, user authentication, and ride search functionalities, which improved the speed and reliability of server responses. I also worked on error handling and middleware logic, ensuring the backend remained secure, optimized, and stable under different use cases.

4.3 Database Structuring and Optimization

I was involved in designing and structuring MongoDB schemas to support efficient data management for users, rides, routes, and booking history. I optimized frequently used queries to reduce data processing time and improve response speed. This contribution significantly enhanced the system's performance and ensured smooth storage and retrieval of ride-related data.

4.4 Collaboration and Team Workflow

Throughout the internship, I worked closely with mentors and fellow trainees using tools like GitHub, Trello, and Slack for coordination and version control. I participated in code reviews, module-wise discussions, and problem-solving sessions that improved my understanding of structured development workflows. This collaboration ensured consistent progress and alignment with project goals.

4.5 Problem Solving and Process Improvement

During the internship, I proposed several improvements to streamline the development process, such as optimizing route comparison logic, refactoring repetitive components, and implementing React Hooks for cleaner state management. My suggestions were implemented in the project and helped enhance code maintainability, reduce complexity, and improve performance.

CHAPTER 5: CONTRIBUTION TO THE ORGANIZATION

The internship at QSpiders was a transformative experience that helped bridge the gap between academic learning and real-world software development. Over the three-month period, I gained valuable technical, professional, and interpersonal skills that contributed significantly to my overall growth as a Computer Science Engineer.

Working on the AI-Based Carpool System gave me hands-on exposure to modern web technologies, full-stack application development, real-time communication tools, and practical implementation of MERN stack concepts. This chapter summarizes the key areas of learning and development achieved during my internship.

5.1 Technical Skills Gained

During my internship, I enhanced my technical knowledge across multiple domains of MERN Full Stack Development. The key technical skills I gained include:

- ❖ **Frontend Development:**

Built and integrated RESTful APIs using Node.js and Express.js, handling authentication, routing, middleware, and real-time data exchange efficiently.

- ❖ **Backend Development:**

Built and integrated RESTful APIs using Node.js and Express.js, handling authentication, routing, middleware, and real-time data exchange efficiently.

- ❖ **AI and Route Optimization:**

Implemented AI-based logic for comparing routes, calculating distances using OpenRouteService, and generating optimized matches between riders and drivers.

- ❖ **Database Management:**

Learned to design and structure MongoDB collections for rides, users, and booking data; optimized queries and improved data retrieval efficiency.

- ❖ **Version Control and Collaboration:**

Used Git and GitHub for source code management, branch handling, pull requests, and collaborative development across modules.

- ❖ **Real-Time Communication:**

Gained experience implementing Socket.io for live ride updates, driver movement tracking, and instant event handling.

- ❖ **Debugging and Optimization:**

Improved debugging efficiency using Chrome DevTools, Postman, and MongoDB Compass, optimizing both frontend responsiveness and backend API performance.

5.2 Professional and Soft Skills Development

The internship also helped me enhance my professional and interpersonal capabilities, preparing me for real-world industry scenarios.

❖ **Teamwork and Collaboration:**

Collaborated effectively with mentors and peers, sharing progress, resolving queries, and coordinating development tasks through tools like Slack, Trello, and GitHub.

❖ **Communication Skills:**

Improved written and verbal communication by participating in discussions, delivering updates, documenting modules, and clarifying technical concepts.

❖ **Time Management:**

Learned to balance multiple modules efficiently within tight practice cycles, prioritizing tasks based on deadlines and complexity.

❖ **Problem-Solving and Critical Thinking:**

Resolved issues during integration, testing, and API implementation by applying analytical thinking, debugging strategies, and structured troubleshooting.

❖ **Adaptability and Learning Mindset:**

Adapted quickly to new frameworks, technologies, and development environments, demonstrating flexibility and curiosity in learning.

5.3 Overall Development

By the end of the internship, I became more confident in handling full-stack development processes from UI creation and API integration to database modeling and algorithmic design. Exposure to real-time tasks, code reviews, and structured practice improved my technical expertise, discipline, and software engineering maturity.

The internship experience at QSpiders not only strengthened my foundation in MERN development but also encouraged me to explore advanced technologies such as intelligent route matching, geospatial APIs, real-time communication systems, and scalable backend architectures. These experiences have inspired me to continue improving my skills and further advance in full-stack development in my career.

CHAPTER 6: SUMMARY AND CONCLUSION

The internship at QSpiders provided a highly enriching and practical experience that helped bridge the gap between academic concepts and real-world software development. Working as a MERN Development Intern gave me the opportunity to engage with modern technologies, industry-standard coding practices, and structured development workflows followed in real software teams.

The project I worked on — an AI-Based Carpool System designed to optimize shared commuting through intelligent route matching, real-time tracking, and automated ride management — exposed me to the complexities of building a scalable and user-friendly full-stack web application. Through this project, I developed strong technical proficiency in React.js, Node.js, Express.js, MongoDB, and real-time technologies such as Socket.io, as well as hands-on exposure to API integration, AI-based route comparison, and map-based services like OpenRouteService and Mapbox.

The internship not only strengthened my technical abilities but also improved my communication, teamwork, time management, and problem-solving skills. Interacting with mentors, participating in module reviews, debugging sessions, and collaborating with other trainees helped me understand how software development life cycles function in real environments — from planning and implementation to testing and optimization.

In conclusion, this internship was a transformative learning experience that shaped both my technical and professional growth. It provided me with the confidence to work on full-stack applications, the mindset to continuously improve, and the adaptability required to excel in the evolving field of software development. The knowledge and experience gained during this internship form a strong foundation for my future career as a Full Stack Developer.

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